

Machine Learning Model Comparison: Analysing Text and Structured Data.

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Abstract – This study investigates the application and performance of machine learning models across three datasets from different domains: crime analytics, supply chain health commodity pricing, and text-based posts. Employing the CRISP-DM methodology, the study began with exploratory data analysis and preparation to ensure reliable modelling and accurate predictions.

The first dataset consists of crime reports from the Los Angeles Police Department (2020–2024). Random Forest and Neural Network models were applied to structured and categorical features, including year of occurrence and crime code, to predict victim age, a variable with limited correlation to the input features. The second dataset focuses on supply chain pricing for health commodity shipments, Decision Tree and Gradient Boosting Machines models predicted freight costs based on features such as quantity, units of measure, and dosage, emphasising interpretability and precision. The third dataset involves posts from non-mental health-related forums—finance, legal advice, and parenting. Text classification was conducted using Glove embeddings to represent textual data, and predictions were made using Random Forest and Logistic Regression models to accurately categorise the posts.

Performance metrics such as R-squared, mean absolute error (MAE), mean squared error (MSE), and F1-score were used to evaluate the models. Results indicated that ensemble methods achieved high R-squared values on structured datasets, while neural networks excelled at capturing complex relationships. Logistic regression effectively handled sparse textual data, achieving strong predictive accuracy. These findings offer valuable insights for selecting machine learning methods suited to structured, numerical, and unstructured datasets across diverse applications.

Index Terms – CRISP-DM methodology, predictive modelling, Random Forest, Gradient Boost, Decision Tree, Neural Network, Logistic Regression.

I. INTRODUCTION

Predictive analytics, which employs advanced statistical techniques [1] and machine learning enables organisations to extract valuable insights from large datasets and make informed predictions [2].

This study investigates the application of machine learning to three predictive tasks: victim age estimation, freight price prediction, and category post classification.

A. Objectives

The primary objective is developing statistical models capable of evaluating and comparing the performance of various machine learning models on these diverse datasets by understanding the unique characteristics of numerical, categorical, and unstructured data, and identifying the most suitable modelling techniques for each task.

B. Methodology

To achieve this goal, the CRISP-DM methodology was employed and involved the following stages: a) Business Understanding, defining the problem, b) Data Understanding, the exploration of data, c) Data Preparation, including data cleaning, data transformation, normalisation, keywords extraction and feature engineering, d) Modelling, involving machine learning model according to numerical and textual nature, and finally e) Evaluation, assess the performance of the models against appropriate metrics [3].

C. Evaluation

The research employs relevant metrics to evaluate the performance of the models, for classification tasks and text prediction metrics like accuracy, precision, recall, and F1-score, and regression tasks, such as predicting victim age and freight price, metrics like mean squared error (MSE) and root mean squared error (RMSE) are employed.

Selecting and interpreting these metrics contributed to understanding the performance of the models and enhanced future work.

II. RELATED WORD

A. Los Angeles Police Department Crimes Report

For Crimes in Los Angeles, several research has been made with data from 2010 to 2020, and from data from 2020 to 2023. The primary approaches are the visualisation of variables and the classification of arrest outcome [4].

First, Naufal Dzaki [5] focuses on an Exploratory Data Analysis about the areas where the crime happened, the most common crimes and weapons, the number of crimes per hour and the count of victims' sex. It generates a narrow age group distribution, which can be considered inappropriate for a regression case because the age is difficult to estimate.

Moreover, crimes are also shown by month and year, in which the year 2023 indicates there were no crimes after July, a comparable situation to the data set of 2020-2024 as the year has not concluded when the data was uploaded and analysed. Meanwhile, Dmytro Iakubovskiy [6] explains, the relationship between the outcome of the crime as a binary output (Arrest / Not Arrest). Analysing the probability of being arrested if the crime was committed throughout the years of study, which criminal code or weapon description was more likely to result in arrest. The approach obtains a correlation between the variables that support the idea of an arrest after the crime was committed.

To estimate age, there have been several studies that help to predict age. For example, Dental age estimation evaluating third molars [7] explains decision tree has been used to predict the age of several population groups providing better performance than regression analysis. Moreover, Multilayer-Perceptron(MLP) can be used for non-linear regression problems [8], as it is efficient at learning complex relationships.

B. Supply Chain Shipment Pricing

Shahrour [9] explores the data set by visualising the main products within the data sets Antiretroviral (ARV) and HIV lab shipments to supported countries. The report focuses on descriptive analysis of variables. F. Xie and W. Yu [10] focus on the discrete variables such as line-item quantity, line-item value, weight (kilograms), and freight cost (USD) to predict logistic costs using K-Nearest Neighbour, Random Forest, and Support Vector Regression machine learning models. The analysis provides the importance of features but only works with a reduced number of features.

C. Non-Mental Health Posts

Natural Language Processing and classification modelling help to classify Reddit posts from BabyBumps and menstruation posts and the primary goal is to create a model that can distinguish which model a post belongs to [11]. It explores a small data set, using a Word Cloud based on word frequency. Logistic Regression and Random Forest help to classify posts, using a pre-trained GloVe embedding due to the pre-defined dense vectors for around every six billion words of English literature [12] provide an application for text classification with large-scale data sets.

III. METHODOLOGY

The CRISP-DM [3] methodology phases applied are:

Business understanding: Defining the project objectives and requirements.

Data Understanding: Accessing the data to describe, explore, and verify its quality.

Data Preparation: Adequate the data for the modelling stage by selecting, cleaning, and formatting data, and feature engineering to find new insights within the data.

Modelling: Selecting and applying appropriate machine learning algorithms.

Evaluation: Assessing the performance of the models.

Deployment: Deploying the final model into a real-world environment.

In this study, deployment has not been applied due to the nature of the research.

A. Los Angeles Police Department Crimes Report

i. Business Understanding

Initially, there were twenty-seven variables available, but only fifteen were numerical. The data has been collected from 2020 until October 2024 and contains 990,293 incidents reported.

The primary objective is to predict the age of the victims and, after that, identify the nature and influence of the variables on the predictable variable through the performance evaluation of different machine learning models. This helps to understand insights into demographic trends in crime analytics.

ii. Data Understanding

Data involves the records of five years of crime reports, which helped to get insights of

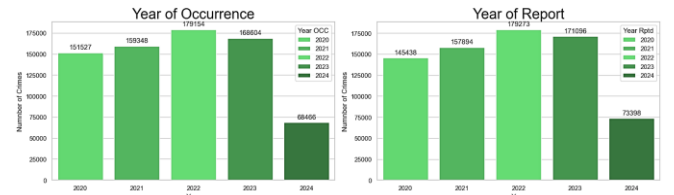


Figure 1: Year of Occurrence vs Year of Report

There is a difference between the occurrence and report date, which in some cases is not the same, as shown in the plot of Figure 1. The year 2024 shows a decrease in crime cases that occur and were reported, but this is subject to the data available as the data for the entire year has not been completed yet.

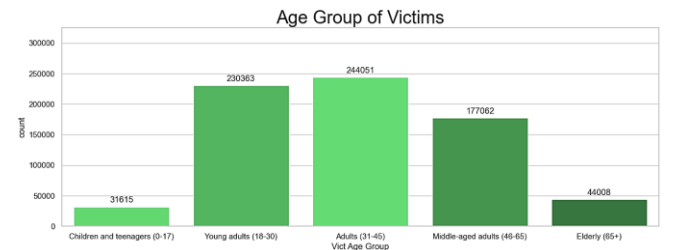


Figure 2. Age Group of Victims

The plot in Figure 2 shows that the most frequent victims are adults between 31 and 45 years old. And the least affected by crime are children and teenagers between 0 and 17 years old.

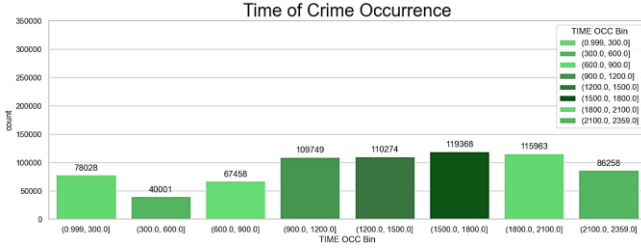


Figure 3: Time of Crime Occurrence

The plot in Figure 3, suggests the time of the day when most of the crimes are likely to occur is between 9:00 am and 9:00 pm, the highest peak between 3:00 pm and 6:00 pm.

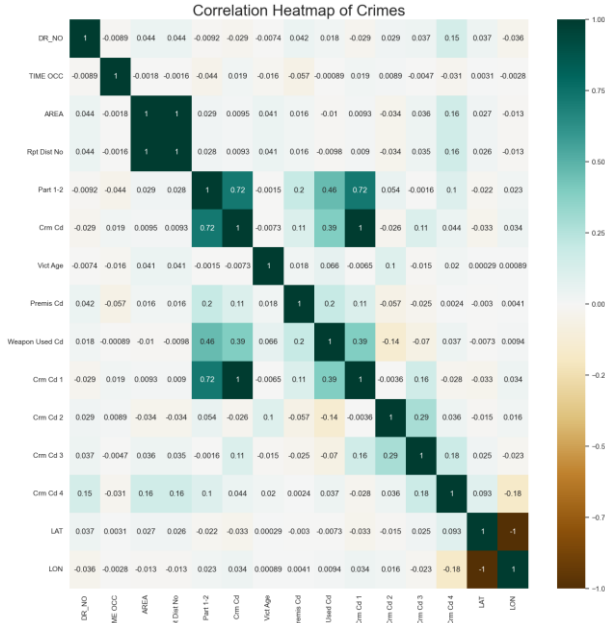


Figure 4: Correlation Heatmap of Crimes

Categorical variables show higher correlations, such as crime codes and the weapon used as shown in Figure 4.

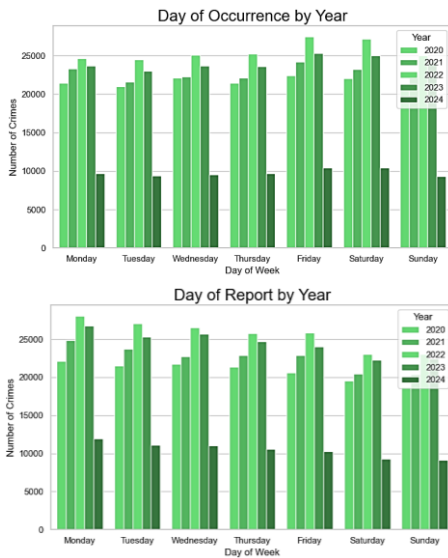


Figure 5: Day of Occurrence and Report per Year Distribution

There is an almost constant distribution of crime occurrence and day of report throughout the week, followed by the most common day of report days from Monday to Wednesday, as shown in the plot in Figure 5.

iii. Data Preparation

Six variables contain at least 50% of missing values, but the nature of these variables is that they depend on others to occur. For example, there is a crime, but no weapon was used, or only a crime involved in the same report. Moreover, other reports involve data that has been omitted, i.e. the age of the victim was recorded, but the sex or descendance was not recorded. Any record that does not include this information has not been considered for the research.

Data cleaning involves transforming categorical into numerical data for processing ends, filling in missing values, and normalisation of data to guarantee the integrity of the data such as Date and Time.

All the variables at this stage are numerical, and the variables selected contain features of geographic data of the crime occurrence and which area was reported, victim features as sex and descendance and time and date of occurrence and report.

iv. Modelling

The clean data set contains 726,931 records, and it was split into 80% for train and 20% for test purposes. For this case, it is necessary to implement machine learning models that recognise non-linear relationships as the Victim age is not related directly to other features.

First, the Decision Tree Regressor was applied to the training data set, allowing it to learn the relationship between the features and the target variable.

Then, for Neural Network with Multilayer Perceptron (MLP) all values needed to be standardized using StandardScaler to ensure optimal performance. The network consists of two hidden layers with ReLU activation and an output layer with no activation function to predict continuous values. The model uses the Adam optimizer, Mean Squared Error as the loss function, and Mean Absolute Error as the evaluation metric. An early stop was implemented to prevent overfitting, with the training process set to stop if the validation loss does not improve for ten consecutive epochs.

For both models, the predictions of the first ten predicted ages were estimated and rounded as a whole value, where MLP proves to have values closer to the actual value than Decision Tree Regressor, as shown in Table 1.

Predictions	1	2	3	4	5	6	7	8	9	10
Actual Value	74	27	24	53	61	29	20	44	20	78
Decision Tree	76	27	23	59	53	25	21	32	30	73
MLP	73	23	27	55	55	26	26	40	26	74

Table 1: First ten predictions vs actual values

v. Evaluation

The metrics obtained from both models are as follows:

Metric	Decision Tree	Neural Network (MLP)
R-Squared	0.8367	0.9128
Mean Absolute Error (MAE)	4.8316	3.7691
Mean Squared Error (MSE) / Loss	39.7108	21.1393
Root Mean Squared Error (RMSE)	6.301	4.6045
Residual Sum of Squares (RSS)	5,773,435	3,082,380

Table 2: Metrics for Crime Data set Regression Models

Table 2 indicates that the Neural Network (MLP) model outperforms the Decision Tree Regressor model across all the metrics provided. MLP explains 7.61% more of the variance in the age of the victim. Also, MAE is lower, meaning its predictions are closer to the true values on average. Moreover, MSE and RMSE indicate better overall prediction accuracy with fewer large errors.

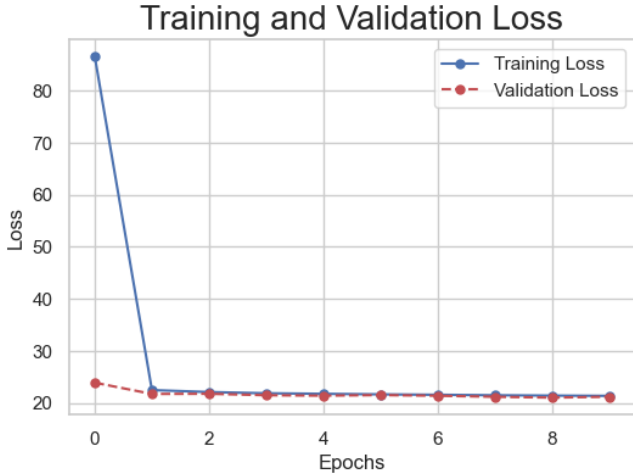


Figure 6: Training and Validation Loss of MLP.

For MLP, as shown in the plot of Figure 6, the training loss decreases rapidly in the initial epochs. This is expected as the model learns to fit the training data better with each iteration. After a few epochs, the training loss continues to decrease, but the validation loss starts to increase which could be a sign of overfitting, that the model is being specialised on the training data and is not generalizing well to new, unseen data.

B. Supply Chain Health Commodity Shipment and Pricing Data Set

i. Business Understanding

The data set identifies Antiretroviral (ARV) and HIV lab shipments to supported countries. [13]. The data provides a wide selection of numerical and categorical features related to delivery dates, prices, dosages, vendors, INCO terms, and detailed information on the commodities themselves.

ii. Data Understanding

Initially, the data set has a variety of numerical and categorical data. But it can be normalised because only the quantity, value, and insurance (USD) of line items are shown and are highly correlated as shown in the plot of Figure 7.

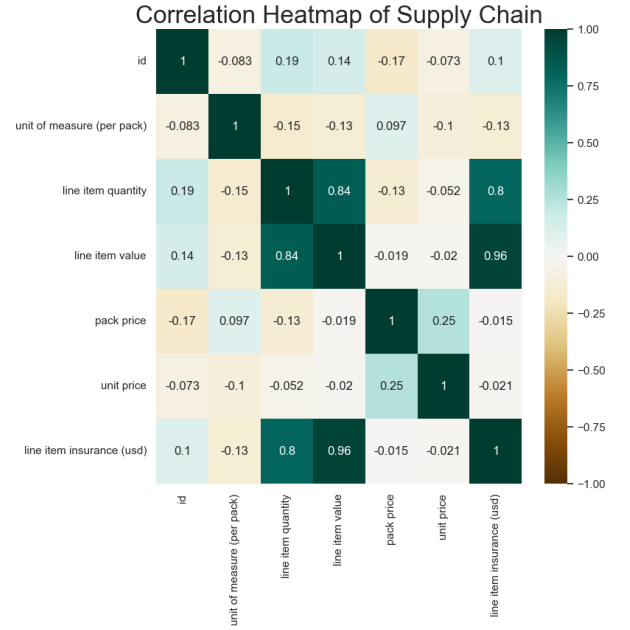


Figure 7: Correlation heatmap of Supply Chain data set.

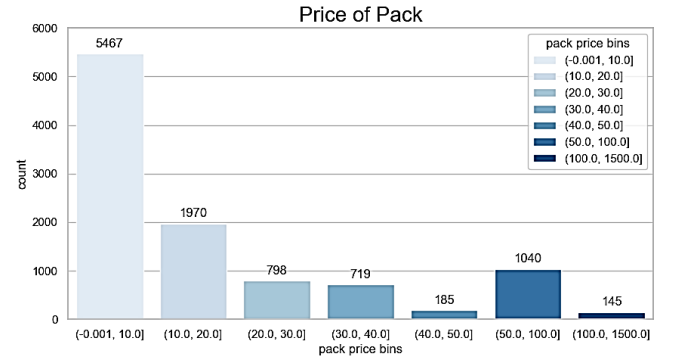


Figure 8: Price Pack distribution in groups.

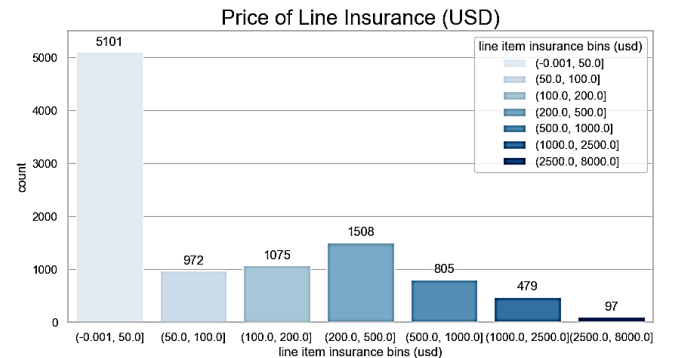


Figure 9: Price of Line Insurance (USD) distribution in groups.

The pack price and price of line insurance(USD) shown in the plot of Figures 8 and 9 indicate there is no normal distribution of the price, most of the pack and line insurance prices are below \$50.00, which means the data is unbalanced.

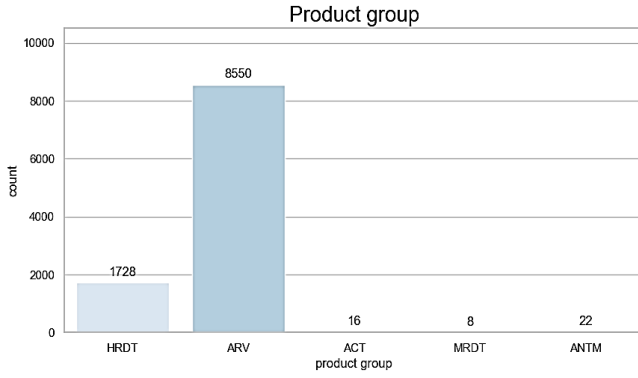


Figure 10: Product Group categories.

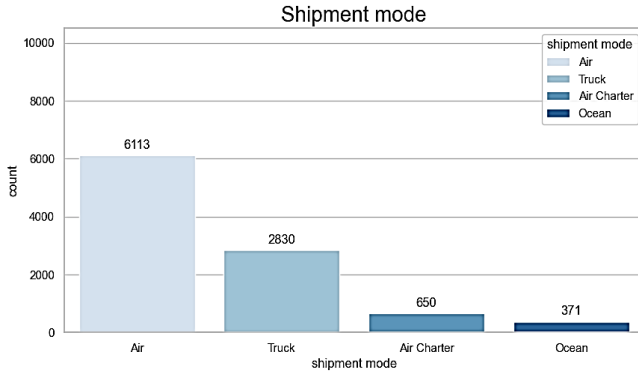


Figure 11: Shipment mode categories.

Also, evaluating the Product group in the plot of Figure 10, most of the products are ARV, and the main Shipment mode is Air as shown in the plot of Figure 11, this could explain the variation in the price of pack and line insurance.

iii. Data Preparation

For this data set, some tasks included are:

- Normalization of variables that require numbers or a date format to ensure consistency (delivery date-related variables, weight in kilograms, freight in USD).
- Handling missing values as non-specified or being dropped from the data set (some freight as zero as could be exempt or included and date not captured) and dropping rows or columns if the missingness is significant and cannot be reliably imputed without biasing the dataset.
- Feature engineering of variables including numerical and categorical data (products with more than one dosage must be separated into dosage and unit measure) to identify new insights.
- Dropping irrelevant features off the data set as not all features in the dataset contribute to meaningful insights or predictions.

After data cleaning, additional variables with meaningful correlations emerge, as shown in Figure 12. These correlations confirm the relevance of these features to the study and their importance for the modelling stage. Identifying these relationships ensures the dataset includes

the most predictive variables, enhancing the accuracy and robustness of the analysis.

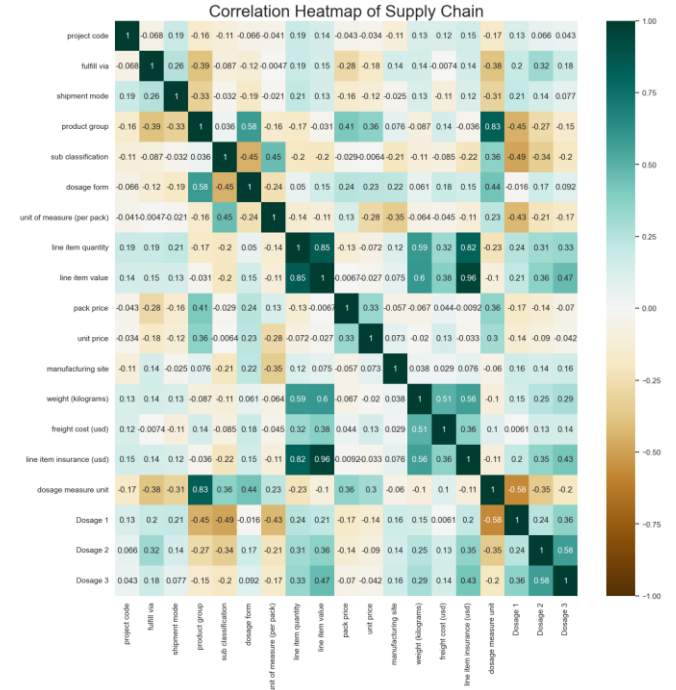


Figure 12: New correlation heatmap of Supply chain variables.

iv. Modelling

The clean data set contains 7,643 records, the split was done into 80% for train and 20% for test. Due to the complexity, size, and robustness of the data set, ensemble methods have been chosen to evaluate their performance.

The first model applied to the train data set is the Random Forest Regressor, combining decision trees to make predictions and it is less prone to overfitting. [14].

The second model applied is Gradient Boosting Machines, which combines multiple weak decision trees sequentially, while each tree learns from the mistakes of the previous ones and captures complex patterns in data. [15].

Predictions	1	2	3	4	5	6	7	8
Actual Value	316.18	1153.68	1188.78	57.6	69.29	10.21	3.59	1.45
Random Forest	350.27	1104.63	1140.1	56.78	70.83	10.03	3.47	1.43
Gradient Boosting Machines	363.89	1140.25	1128.48	57.5	67.22	4.82	4.03	4.51

Table 3: First ten predictions vs actual values

For the first ten values from the data set, the Random Forest Regressor suggests more accurate predictions than the Gradient Boosting Machines model, as shown in Table 3.

v. Evaluation

The metrics obtained from both models are as follows:

Metric	Random Forest	Gradient Boosting Machines
R-Squared	97.39	98.04
Mean Absolute Error (MAE)	33.36	29.32
Mean Squared Error (MSE) / Loss	7038.65	5289.38
Root Mean Squared Error (RMSE)	83.90	72.73
Residual Sum of Squares (RSS)	10,762,094.39	8,087,468.65

Table 4: Metrics for Supply Chain Data Set Regression Models

According to the values observed in Table 4, the Gradient Boosting Machines model outperforms the Random Forest in this specific case. While both models exhibit strong predictive power, achieving high R-squared scores, the R-squared of Gradient Boosting Machines suggests that they effectively capture 0.65% more variability than Random Forest in the data.

This is particularly evident in the lower Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Residual Sum of Squares (RSS), indicating that Gradient Boosting Machines predictions are, on average, closer to the actual values.

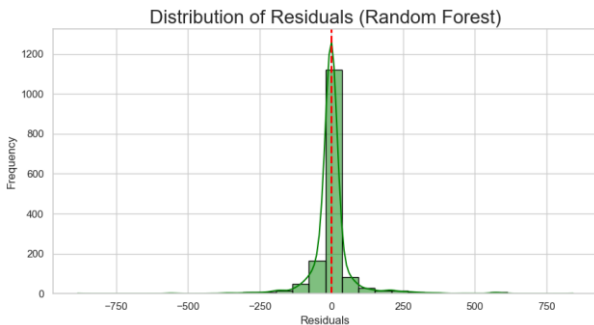


Figure 13: Distribution of Residuals using Random Forest

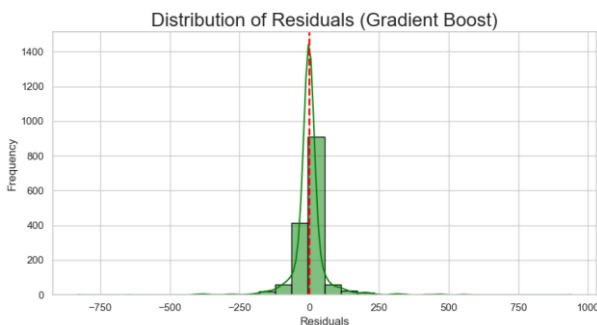


Figure 14: Distribution of Residuals using Gradient Boost

Furthermore, the residuals from the Gradient Boosting Machines model appear to be more normally distributed as shown in the plot of Figure 13 compared to the plot in Figure 12 which shows higher values for Residuals from the Random Forest model.

Additionally, a five-fold cross-validation was applied to both models, and it indicates that the differences between predicted values and actual values are lower for Gradient

Boosting Machines, suggesting that its predictions are, on average, closer to the actual values as shown in Table 5.

Model	Mean CV-RMSE
Random Forest Regressor	108.28
Gradient Boosting Machines	88.15

Table 5: Cross Validation mean RMSE of machine learning models.

C. NON-MENTAL HEALTH POSTS

i. Business Understanding

Reddit contains several categories from twenty-eight different health groups from 2018 to 2020. However, only data from 2019 has been selected and gathered from only three different non-mental health subreddits. The main goal is to build a model to predict and distinguish which category a post belongs to.

ii. Data Understanding

There are three categories of posts, personal finance, parenting, and legal advice in a total of 19,437 posts and each category has a similar number of posts to provide future models the ability to recognise each.

```
text["post"].head()
[23] ✓ 0.0s Python
... 0 Withdrew my entire 401k at 23, how fucked am I...
1 Always ask for a discount on your hospital bil...
2 Affordable full coverage insurance? Sorry if t...
3 Can I afford this car? Should I get it? Monthl...
4 Still have a student bank account but I droppe...
Name: post, dtype: object
```

Figure 15: First five posts of data set.

The posts have been written by users of Reddit and contain grammatical errors, contractions, and symbols that are not processed or need normalisation, as seen in Figure 15.

iii. Data Preparation

Through the Spacy library, the English language model for NLP is loaded, and after some word cloud attempts, a custom set of stop words that are not useful for analysis is defined.



Figure 16: Word Cloud of posts.

After that, a word cloud has been created, as shown in the plot of Figure 16, providing a high-level view of frequently discussed topics, and highlighting key terms.

To identify patterns or groupings in the data without prior knowledge of categories, clustering was applied, to identify the top ten words in each cluster and the result is:

- Cluster 0: car, money, pay, just, account, ampx200b, company, work, job, year.
- Cluster 1: kids, just, old, son, like, daughter, time, child, year, does.
- Cluster 2: credit, card, score, debt, loan, pay, cards, balance, car, student.

It can be recognised that Cluster 0 reflects financial concerns, Cluster 1 represents parenting and family issues, and Cluster 2 suggests credit and debt management identifying patterns and summarising the complexity of the posts into clusters.

iv. Modelling

Two classification models using word embeddings have been trained to evaluate their performance. The models applied are Random Forest Classifier (RF) and Logistic Regression (LR).

Also, GloVe embeddings converted text posts into numerical vectors that represent the semantic meaning of the words. After that, the vectors are fitted to each of the machine-learning models. The model uses one pre-trained GloVe embedding (glove.6B.300d.txt) with a dimensionality of three hundred [16]. For each text post, the vectors for all words in the post are averaged to create a single vector that represents the post.

The data has been split into 80% and 20% for train and test sets respectively, and the target variable is “subreddit_id”, as follows “0” is legal advice, “1” represents parenting and “2” reflects personal finance.

The Random Forest has been trained using one thousand decision trees, and for the Logistic Regression, the maximum number of iterations assigned is one thousand, to ensure it has enough iterations to intersect.

v. Evaluation

The metrics obtained through the modelling stage are as follows:

Random Forest	precision	recall	f1-score
accuracy	0.884774	0.884774	0.884774
0	0.849741	0.851632	0.850685
1	0.899096	0.929907	0.914242
2	0.908189	0.874204	0.890872
macro avg	0.885675	0.885247	0.885267
weighted avg	0.884922	0.884774	0.884657

Table 6: Random Forest for Posts data set metrics

Logistic Regression	precision	recall	f1-score
accuracy	0.915895	0.915895	0.915895
0	0.887097	0.897626	0.892330
1	0.945004	0.950156	0.947573
2	0.917275	0.900478	0.908799
macro avg	0.916459	0.916087	0.916234
weighted avg	0.915969	0.915895	0.915894

Table 7: Logistic Regression for Posts data set metrics

Logistic Regression outperforms Random Forest in every metric: accuracy, precision, recall, and F1-score as shown in Tables 5 and 6. In terms of overall accuracy, LR improves by 3.1% more than RF. This suggests that the LR model is better at correctly classifying posts.

LR also outperforms Random Forest in terms of recall. The difference is most pronounced in Class 1 (around 2% better). This shows that Logistic Regression is better at identifying relevant posts, particularly in Class 1.

IV. CONCLUSIONS AND FUTURE WORK

A. Los Angeles Police Department Crimes Report

i. Key Findings:

Predicting crime across different areas presents a significant challenge due to the complex and non-linear relationships between crime and its influencing factors. Variables such as geographic location, crime classifications, weapons used, and modus operandi are crucial for identifying crime patterns, but they are often disproportionate or inaccurate.

The variables do not suggest a high or linear correlation with the age of victims, indicating that the age of victims does not directly depend on crime-related data.

As a result, predicting the age of crime victims requires machine learning models capable of handling these non-linear dynamics, such as the Decision Tree Regressor and the Neural Network with a Multilayer Perceptron (MLP). These models are particularly effective in predicting victim age. MLP proves to be more suitable than the Decision Tree Regressor due to its ability to model complex, non-linear relationships through its multiple layers and interconnected nodes, making it well-suited for uncovering hidden patterns in large-scale crime datasets.

ii. Implications and Limitations.

For the Decision Tree regressor, hyperparameter tuning through Grid Search CV [17] could limit the maximum depth of the tree and increase the value of the minimum sample split to make the tree more conservative and then applying pruning techniques can decrease overfitting within the model, making it more capable of generalising.

For the Neural Network with Multi-Layer Perceptron (MLP), the experimentation with the number of hidden layers

and the number of neurons in each layer can help the model capacity to learn more complex patterns, but not exceeding the depth as it can lead to overfitting.

The inclusion of demographic features can help to discover additional patterns and identify the areas and age groups more prone to be affected by crime, making the Los Angeles Police Department able to develop targeted action plans to ensure safety and maintain order in high-crime areas.

B. Supply Chain Health Commodity Shipment and Pricing Data Set.

iii. Key Findings.

The data for health commodity shipment and pricing is unbalanced, as most of the product prices are relatively very low (under \$50). This indicates that freight price (USD) is influenced by several variables. For instance, variables such as weight exhibit a high correlation with freight costs. Additionally, other variables like dosage and product group also have significant interdependencies, influencing each other in complex ways.

iv. Implications and Limitations.

Hyperparameter tuning through Grid Search CV in both, Random Forest Regressor and Gradient Boosting Machines can result in capturing more complex patterns within the freight price (USD), by experimenting with different numbers of estimators to determine the number of trees in the forest. Limiting the depth of individual trees helps prevent overfitting. Also, determining the minimum sample leaf size at the end of the tree can help with pruning and reduce overfitting [18].

Moreover, the Recursive Feature Elimination (RFE) method can be applied to the Random Forest, to remove irrelevant or highly correlated features that do not contribute to the predictive capacity of the model until the model reaches a reasonable number of features as it can manage high-dimensional data sets [19]. It must be considered that the computational resources needed to analyse the large-scale data set increase with the proposed improvements.

C. Non-Mental Health Posts

i. Key Findings.

Providing enough posts helps to develop machine learning models capable of recognising patterns in text and classifying according to personal finance, legal advice and parenting. In which the Logistic Regression model outperformed the Random Forest across all metrics but proved to be capable of classifying each category.

ii. Implications and Limitations.

A larger number of posts from each of the categories or more diverse Reddit categories, such as mental health, can enhance the power of the model. Expanding the dataset makes the model rich in a variety of linguistic patterns, context and

semantic relationships. Furthermore, adding more recent posts can evaluate the content changes pattern throughout the years and the topics that were defined providing a temporal depth to the analysis, although it can require significantly higher computational resources, and it can introduce noise or inconsistencies to the model requiring advanced preprocessing techniques to maintain data quality.

Instead of using pre-trained embeddings, consider fine-tuning GloVe embeddings, adapting the pre-trained model on a larger dataset of Reddit posts to capture the platform's unique linguistic patterns that improve performance and make it more efficient and accessible [20].

In this context, more sophisticated models like transformers like BERT which reads text in both directions (left-to-right and right-to-left) simultaneously, allowing it to understand the full context of a word., designed by Google [21], which improves the understanding of context and context-dependent word representations providing a deeper semantic understanding of Reddit posts.

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